

Marketing A/B Testing Analysis: The Effect of Advertisement on Customer Conversion Rate

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Abstract

This report presents an A/B testing analysis of a digital marketing campaign, examining whether exposure to a commercial advertisement (*ad*) increases conversion rate compared to a public service announcement (*psa*) across 588,101 users. Both Frequentist and Bayesian approaches are applied to a binary conversion outcome (*converted*) and subgroup analyses by day and hour of peak ad exposure.

Frequentist results show a statistically significant difference in conversion rate ($\chi^2(1) = 54.006$, $p < 0.001$), with the *ad* group achieving 2.55% compared to 1.79% for *psa*. Bayesian posterior analysis confirms a 100% probability that *ad* yields a higher conversion rate, with an estimated advantage of 0.59%–0.94%. Subgroup analysis identifies Monday and the afternoon hours as the highest-performing conditions. All effect sizes remain negligible, though the consistency across both methods supports the effectiveness of commercial advertisement exposure.

Based on these findings, advertisers are recommended to prioritize ad delivery on Monday and during afternoon hours, with increased frequency targeting highly engaged users.

1 Introduction

Understanding the effectiveness of digital advertisement is critical for optimizing marketing strategies and maximizing return on investment. While traditional metrics such as click-through rates provide surface-level insights, conversion rate, the proportion of users who complete a purchase after exposure, serves as a more direct measure of advertisement effectiveness.

This report presents an A/B testing analysis of a digital marketing campaign, examining whether exposure to a commercial advertisement (*ad*) leads to a higher conversion rate compared to a public service announcement (*psa*). The dataset contains 588,101 user records, with users randomly assigned to one of two experimental groups.

Both Frequentist and Bayesian approaches are applied to evaluate the primary binary outcome (*converted*) and to examine conversion patterns across days and hours of peak ad exposure. The Frequentist approach determines whether observed differences are statistically significant, while the Bayesian approach provides posterior probabilities and credible intervals for a more interpretable conclusion.

2 Data

2.1 Dataset Overview

The dataset contains 588,101 user records sourced from Kaggle, collected from a randomized A/B experiment conducted on a digital marketing campaign. Users were randomly assigned to one of two experimental groups: *psa* (control, $n = 23,524$) and *ad* (treatment, $n = 564,577$). No missing values were detected across all variables. The primary outcome variable is *converted*, a binary indicator representing whether a user made a purchase after being exposed to the advertisement.

2.2 Feature Description

The dataset contains 6 variables: 1 identifier (*user.id*, excluded from analysis), 1 grouping variable (*test.group*), 1 behavioral metric (*total.ads*), 1 binary outcome variable (*converted*), and 2 temporal variables (*most.ads.day*, *most.ads.hour*). *test.group* indicates the experimen-

tal group assigned to each user, either *psa* (control) or *ad* (treatment). *total.ads* records the total number of advertisements viewed by each user. *converted* is a logical variable indicating whether a user completed a purchase. *most.ads.day* and *most.ads.hour* record the day and hour of peak ad exposure for each user, respectively.

3 Exploratory Data Analysis

3.1 Summary Statistics

The dataset contains 588,101 complete observations with no missing values detected. The two experimental groups are imbalanced by design: *psa* contains 23,524 users (4.0%) and *ad* contains 564,577 users (96.0%). The overall conversion rate is 2.52%, with *ad* achieving 2.55% and *psa* achieving 1.79%. *total.ads* ranges from 1 to 2,065 with a mean of 24.82 and a median of 13, indicating a heavily right-skewed distribution.

3.2 Conversion Rate by Test Group

Figure 1 shows the conversion rates for both experimental groups. The *ad* group achieves a higher conversion rate (2.55%) compared to the *psa* group (1.79%), indicating a descriptive difference of 0.76 percentage points.

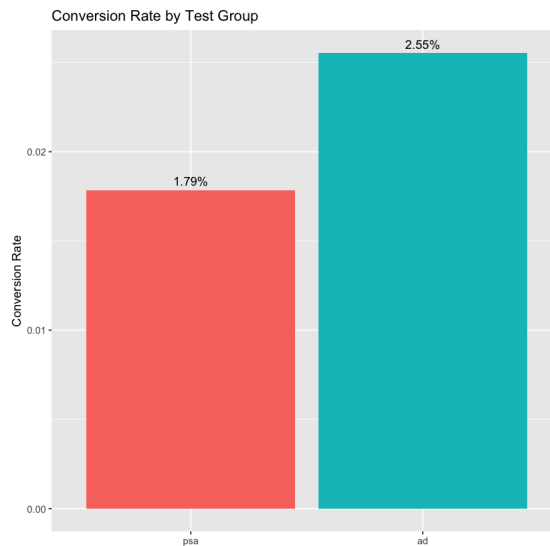


Figure 1: Conversion Rate by Test Group

3.3 Conversion Rate by Day and Hour

As shown in Figure 2, Monday yields the highest conversion rate in the *ad* group, while Thursday, Friday, and Saturday rank lower. Figure 3 indicates that conversion peaks during 14:00–20:00, with hour 16 reaching the highest rate.

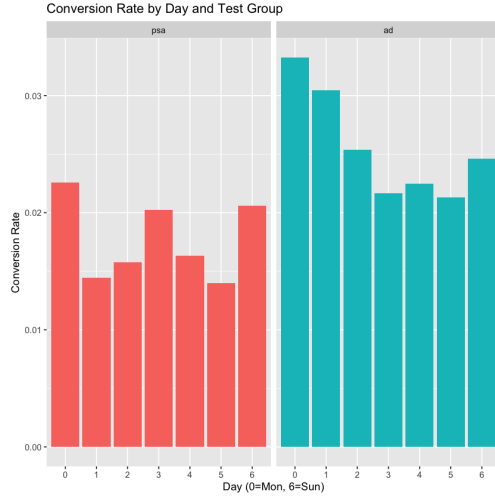


Figure 2: Conversion Rate by Day

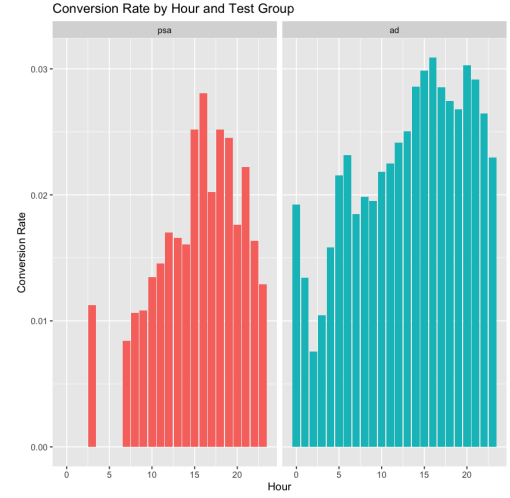


Figure 3: Conversion Rate by Hour

Figure 4 and Figure 5 present the conversion rate and sample size heatmaps for the *ad* group. Notably, some cells with elevated conversion rates correspond to cells with substantially fewer observations, suggesting that sample size varies considerably across day–hour combinations.

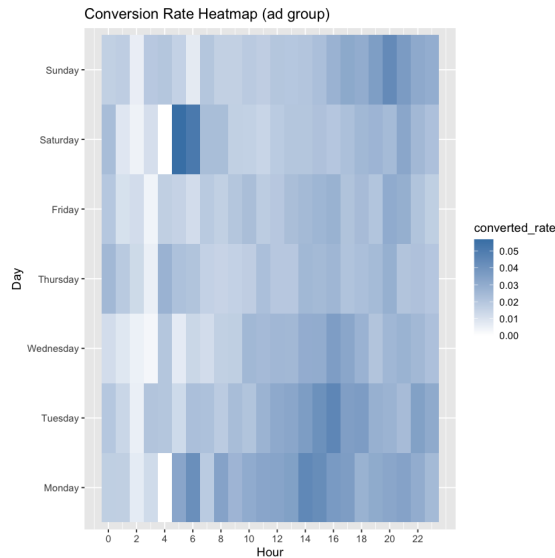


Figure 4: Conversion Rate Heatmap

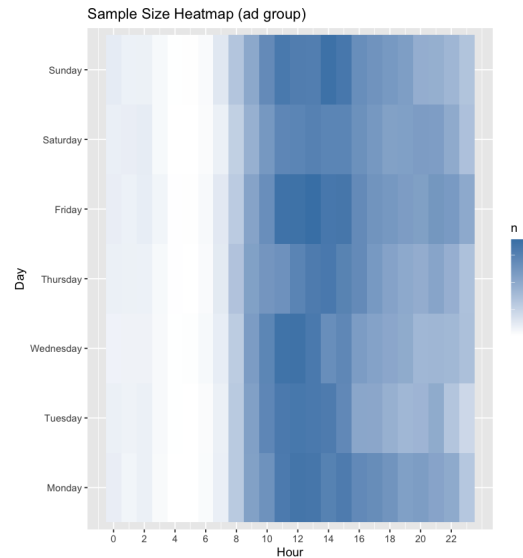


Figure 5: Sample Size Heatmap

3.4 Outlier Detection

Figure 6 visualizes the outlier threshold across both groups on a log scale. To assess the presence of extreme values in *total.ads*, the IQR method was applied with a threshold of $3 \times \text{IQR}$, yielding an upper bound of 96, with 24,825 observations (4.22%) exceeding this threshold.

Outlier users exhibit substantially higher conversion rates (17.18% for *ad* and 11.51% for *psa*) compared to non-outlier users (1.91% and 1.31%, respectively).

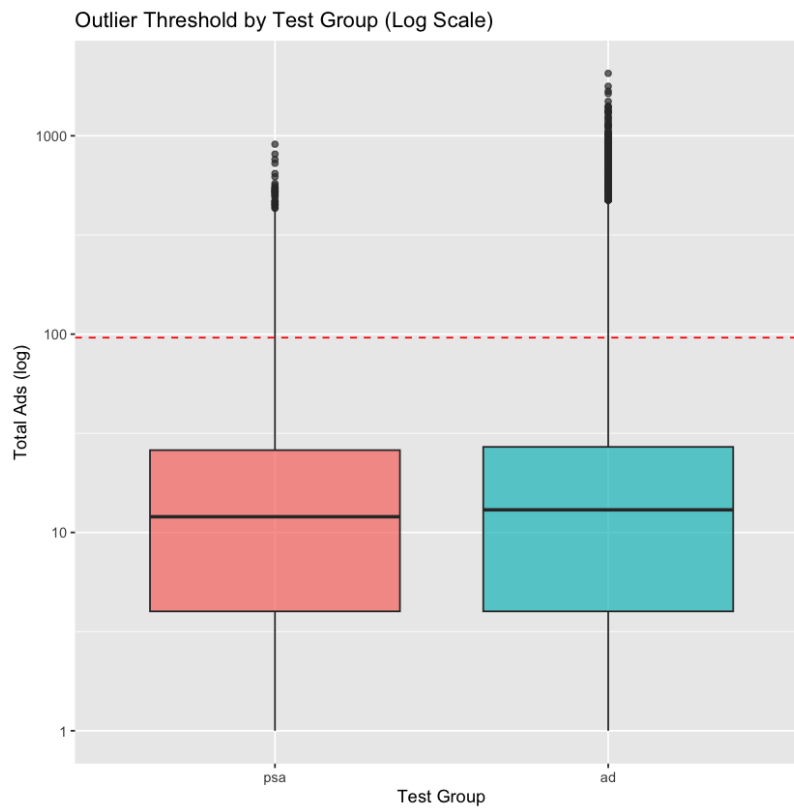


Figure 6: Total Ads Distribution by Group (Log Scale). The dashed red line indicates the $\text{IQR} \times 3$ outlier threshold.

4 Methodology

4.1 Frequentist Approach

Mann-Whitney U Test The Mann-Whitney U test was applied to compare *total.ads* between the two groups. As a non-parametric test, it does not assume normality and is therefore appropriate for the right-skewed distribution of ad exposure counts.

Two-Sample Proportion Test The two-sample proportion test was applied to evaluate whether advertisement exposure is associated with a higher conversion rate. The null hypothesis states that the conversion rate is independent of group assignment.

Chi-Square Test The Chi-square test of independence was applied to evaluate whether conversion rate varies across days and hours within the *ad* group.

4.2 Bayesian Approach

Beta-Binomial Model For binary conversion outcomes, a Beta-Binomial conjugate model was applied with an uninformative prior $\text{Beta}(1, 1)$, yielding the posterior:

$$p \mid \text{data} \sim \text{Beta}(\alpha + \text{successes}, \beta + \text{failures})$$

Gamma-Poisson Model For *total.ads*, a Gamma-Poisson conjugate model was applied with prior $\text{Gamma}(1, 1)$, yielding the posterior:

$$\lambda \mid \text{data} \sim \text{Gamma}\left(\alpha + \sum x_i, \beta + n\right)$$

4.3 Evaluation Metrics

Frequentist Frequentist tests are evaluated using p-value and effect size: rank-biserial r for the Mann-Whitney U test, Cohen’s h for the two-sample proportion test, and Cramér’s V for Chi-square tests.

Bayesian For Bayesian models, Monte Carlo sampling ($n = 100,000$) was used to estimate the following metrics:

- $P(ad > psa)$: posterior probability that *ad* outperforms *psa*
- **95% HDI**: the Highest Density Interval representing the most credible range of the difference between groups

5 Frequentist Results

5.1 Randomization Check (total.ads)

A randomization check confirms that the two groups received comparable ad exposure (mean: 24.76 vs. 24.82), with no meaningful difference detected by the Mann-Whitney U test ($r = 0.0086$, small), supporting the validity of the experimental design.

5.2 Primary Analysis (converted)

The two-sample proportion test reveals a statistically significant but negligible difference in conversion rate between the two groups ($\chi^2(1) = 54.006$, $p < 0.001$, Cohen's $h = 0.053$). The *ad* group achieves 2.55% compared to 1.79% for *psa*.

5.3 Subgroup Analysis (ad group only)

5.3.1 Conversion Rate by Day

As shown in Table 1, Monday achieves the highest conversion rate (3.32%), while Saturday records the lowest (2.13%). This pattern is supported by the Chi-square residuals, where Monday shows the largest positive residual (+13.92) and Saturday the largest negative (-7.45).

Conversion Rate by Day (ad group)			
Day	n	Conversion Rate	Std. Residual (TRUE)
Monday	83,571	3.32%	+13.92
Tuesday	74,572	3.04%	+8.36
Wednesday	77,418	2.54%	−0.33
Thursday	79,077	2.16%	−6.88
Friday	88,805	2.25%	−5.75
Saturday	78,802	2.13%	−7.45
Sunday	82,332	2.46%	−1.66

Table 1: Conversion Rate and Standardized Residuals by Day (ad group)

The Chi-square test confirms a statistically significant but negligible difference across days ($\chi^2(6) = 412.79$, $p < 0.001$, Cramér’s $V = 0.027$), with conversion rates ranging from 2.13% to 3.32%. Pairwise tests with Bonferroni correction indicate that Monday differs significantly from all other days ($p \leq 0.034$), while Thursday, Friday, and Saturday show no significant differences among themselves ($p = 1.000$).

5.3.2 Conversion Rate by Hour

Table 2 presents the conversion rates for the top 10 performing hours within the *ad* group. Hour 16 achieves the highest conversion rate (3.09%, residual = +6.34), while hour 13 records the lowest among the top 10 (2.51%), a difference of 0.58 percentage points.

Conversion Rate by Hour (ad group, Top 10)			
Hour	n	Conversion Rate	Std. Residual (TRUE)
13	45,485	2.51%	−0.64
14	43,779	2.86%	+3.96
15	42,855	2.98%	+5.57
16	35,963	3.09%	+6.34
17	33,605	2.85%	+3.43
18	31,052	2.75%	+2.12
19	29,169	2.68%	+1.35
20	27,846	3.03%	+4.94
21	28,895	2.92%	+3.86
22	25,515	2.65%	+0.91

Table 2: Conversion Rate and Standardized Residuals by Hour, Top 10 (ad group)

The Chi-square test confirms a statistically significant but negligible difference across hours ($\chi^2(23) = 403.33$, $p < 0.001$, Cramér’s $V = 0.0267$). Pairwise tests with Bonferroni correction show that hours 15, 16, 20, and 21 differ significantly from hour 13 ($p \leq 0.034$), while hours 14, 17, 18, 19, and 22 show no significant differences from hour 13.

6 Bayesian Results

6.1 Randomization Check (total.ads: Gamma-Poisson)

A randomization check using the Gamma-Poisson model confirms comparable ad exposure between groups ($P(ad > psa) = 97.19\%$, 95% HDI: $[-0.0013, 0.1284]$), with the HDI including zero, indicating no meaningful difference between groups.

6.2 Primary Analysis (converted: Beta-Binomial)

The Beta-Binomial model yields a posterior probability of $P(ad > psa) = 1.0000$, with a 95% HDI of $[0.0059, 0.0094]$ that excludes zero, confirming that the *ad* group achieves a higher conversion rate with near certainty. The posterior means are 2.55% for *ad* and 1.79% for *psa*. Figure 7 shows the posterior distributions of the conversion rate for both groups, with the *ad* distribution shifted clearly to the right.

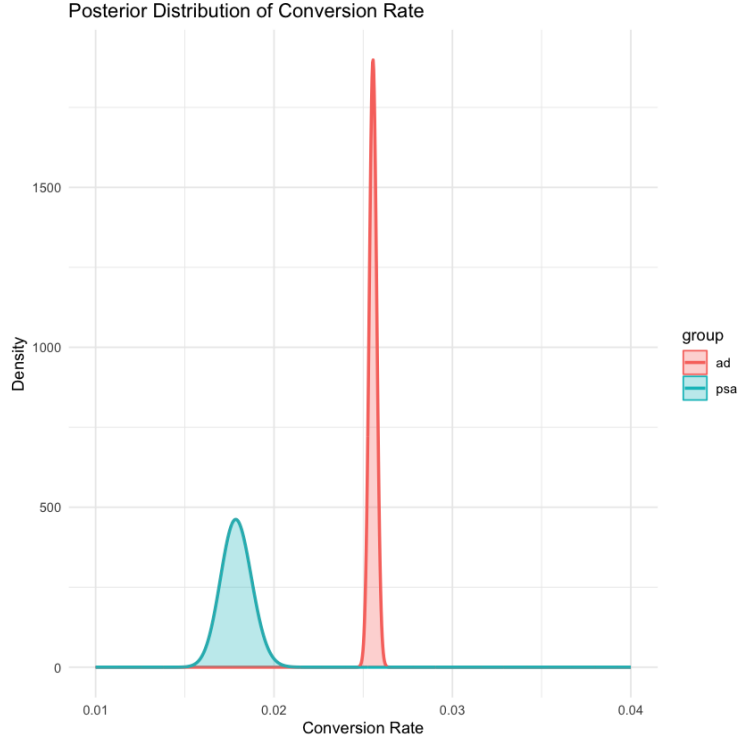


Figure 7: Posterior Distribution of Conversion Rate (Beta-Binomial)

6.3 Subgroup Analysis (ad group only)

6.3.1 Conversion Rate by Day

As shown in Table 3, Monday achieves the highest posterior mean (3.33%), while Saturday records the lowest (2.13%). The HDIs of Monday [3.20%, 3.45%] and Saturday [2.03%, 2.23%] do not overlap, confirming that the difference is credible rather than due to sampling variability.

Posterior Conversion Rate by Day (ad group)			
Day	Posterior Mean	HDI Lower	HDI Upper
Monday	3.33%	3.20%	3.45%
Tuesday	3.05%	2.92%	3.17%
Wednesday	2.54%	2.43%	2.65%
Thursday	2.17%	2.07%	2.27%
Friday	2.25%	2.15%	2.34%
Saturday	2.13%	2.03%	2.23%
Sunday	2.46%	2.36%	2.57%

Table 3: Posterior Conversion Rate by Day (ad group)

Pairwise posterior comparisons show that Monday outperforms all other days with near certainty ($P \geq 0.9993$, all 95% HDIs exclude zero). Figure 8 shows the posterior distributions for Monday, Wednesday, and Saturday, representing the high, mid, and low performing days.

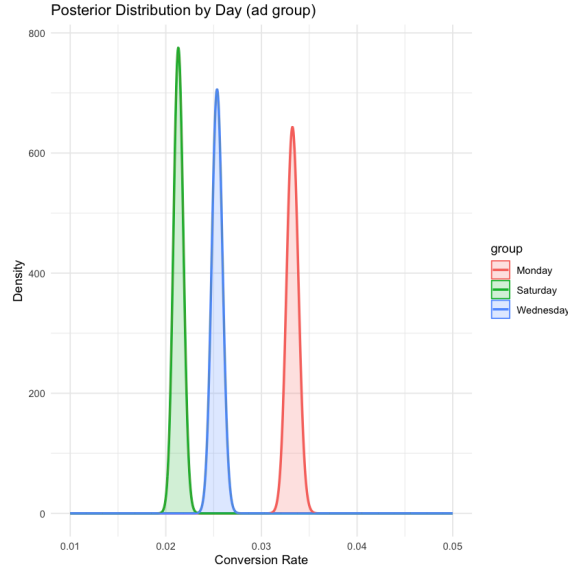


Figure 8: Posterior Distribution of Conversion Rate by Day (Monday, Wednesday, Saturday)

6.3.2 Conversion Rate by Hour

As shown in Table 4, hour 16 achieves the highest posterior mean (3.09%), while hour 13 records the lowest (2.51%) among the top 10 performing hours. The HDIs of hour 16 [2.91%, 3.27%] and hour 13 [2.37%, 2.65%] do not overlap, confirming a credible difference.

Posterior Conversion Rate by Hour (ad group, Top 10)				
Hour	Posterior Mean	HDI Lower	HDI Upper	
13	2.51%	2.37%	2.65%	
14	2.86%	2.70%	3.02%	
15	2.99%	2.83%	3.15%	
16	3.09%	2.91%	3.27%	
17	2.86%	2.68%	3.04%	
18	2.75%	2.57%	2.93%	
19	2.68%	2.50%	2.87%	
20	3.03%	2.83%	3.23%	
21	2.92%	2.72%	3.11%	
22	2.65%	2.45%	2.85%	

Table 4: Posterior Conversion Rate by Hour, Top 10 (ad group)

Pairwise posterior comparisons show that hour 16 outperforms hour 13 with near certainty ($P = 1.0000$, HDI: $[0.0036, 0.0082]$), and also shows credible advantages over hours 18, 19, and 22 (HDI excludes zero). However, no meaningful difference is found between hour 16 and hours 14, 15, 17, 20, and 21 (HDI includes zero), suggesting these top-performing hours achieve comparable conversion rates. Figure 9 shows the posterior distributions for hours 13, 14, and 16.

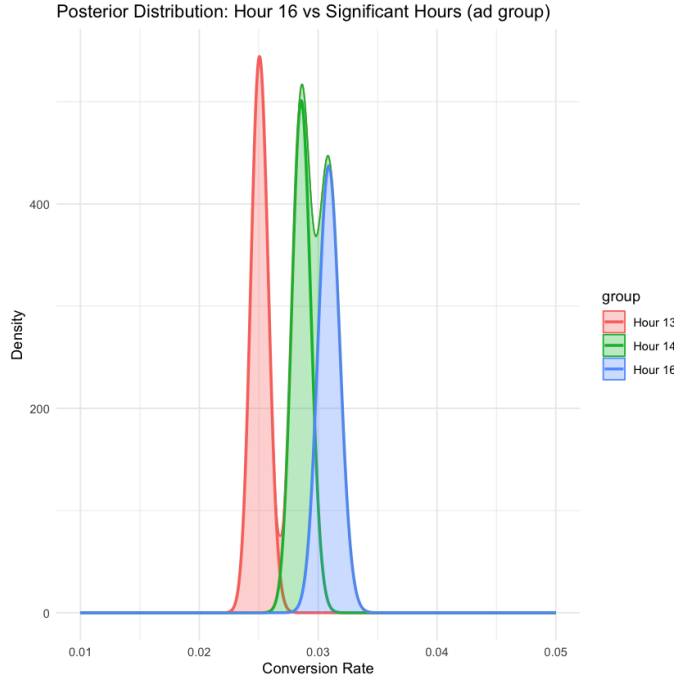


Figure 9: Posterior Distribution of Conversion Rate by Hour (Hour 13, 14, 16)

7 Overall Comparison

Both Frequentist and Bayesian approaches consistently confirm that advertisement exposure significantly increases conversion rate. The Frequentist two-sample proportion test yields $p < 0.001$, while the Bayesian Beta-Binomial model estimates $P(ad > psa) = 1.0000$ with a 95% HDI of $[0.0059, 0.0094]$ excluding zero. However, both approaches agree that the practical effect is negligible (Cohen’s $h = 0.053$).

For subgroup analysis, both approaches identify Monday and the afternoon hours as the highest-performing conditions, with hour 16 achieving the peak conversion rate. All effect sizes remain negligible, consistent with the large sample size inflating statistical significance.

8 Discussion & Conclusion

8.1 Key Findings

Both Frequentist and Bayesian analyses consistently confirm that advertisement exposure has a statistically significant positive effect on conversion rate. The *ad* group achieves a conversion rate of 2.55%, compared to 1.79% for the *psa* group, with near-certain posterior probability ($P = 1.0000$). Despite statistical significance, all effect sizes remain negligible, reflecting a common pattern in large-sample A/B tests where even trivial differences reach significance.

8.2 Business Recommendations

Based on the analysis, advertisement exposure is recommended to be prioritized on **Monday** and during **afternoon hours**, particularly at **hour 16**, where conversion rates are consistently highest.

Users with high ad exposure ($total.ads > 96$) exhibit substantially higher conversion rates (17.18%), suggesting that sustained advertisement exposure has a cumulative effect on purchase likelihood. Advertisers may therefore benefit from increasing ad frequency for engaged users rather than broadening reach to new audiences.

8.3 Limitations

Dataset Limitations

The dataset does not include contextual variables such as user demographics, device type, or advertisement content, which may confound the relationship between ad exposure and conversion. Additionally, it is unclear whether *total.ads* represents unique advertisement impressions or repeated exposures to the same advertisement, limiting the interpretability of the cumulative exposure effect.

Evaluation Metrics

The analysis relies solely on *converted* as the primary outcome metric. While conversion captures whether a purchase occurred, it does not reflect revenue per user or customer lifetime value, which would provide a more comprehensive evaluation of advertisement effectiveness. Furthermore, all effect sizes remain negligible despite statistical significance, suggesting that the practical impact of the findings should be interpreted with caution.

References

- [1] Vázquez, F. *Marketing A/B Testing* [Dataset]. Kaggle.
<https://www.kaggle.com/datasets/faviovaz/marketing-ab-testing>